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HR 6094: A YOUNG SOLAR-TYPE, SOLAR-METALLICITY BARIUM DWARF STAR[1]

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ABSTRACT

The young solar-type star HR 6094 is found to be a barium dwarf, overabundant in the *s*-process elements as well as deficient in C. It is a member of the solar-metallicity, 0.3 Gyr old Ursa Major kinematical group. Measurements of radial velocity and ultraviolet flux do not support the attribution of such abundance anomalies to an unseen degenerate companion. A common proper motion, $V = 10$, DA white dwarf (WD), located 5360 AU away, however, strongly supports the explanation of the origin of this barium star by the process of mass transfer in a binary system, in which the secondary component accreted matter from the primary one (now the WD) when it was an asymptotic giant branch (AGB) star self-enriched in the *s*-process elements. The membership in the UMa group of *another s*-process–rich and C-deficient star, HR 2047, suggests that these stars could have formed a multiple system in the past, which was disrupted by the mass-loss episode of the former AGB star. Their [C/Fe] deficiency could be explained by the action of the hot-bottomed envelope burning process in the late AGB, thereby reconverting it from a C-rich to an O-rich star, depleting C while enriching its envelope with Li and neutron capture elements. This is the first identification of the barium phenomenon in a near–zero-age star, besides being the first barium system in which the remnant of the late AGB star responsible for the heavy-element enrichment may have been directly spotted.

Subject headings: stars: abundances — stars: AGB and post-AGB — stars: chemically peculiar — white dwarfs

1. INTRODUCTION

There is near-consensus that the most promising explanation for the abundance anomalies of the barium stars is the mass transfer scenario of McClure (1983), which invokes membership in a binary system. In this scenario, the former, more massive primary star evolves as a thermally pulsing asymptotic giant branch (TP-AGB) star and enriches its He-burning envelope with the products of neutron capture processes. These are later transferred to the surface by the "third dredge-up" (Iben & Renzini 1983) convective episode, where they are expelled by mass-loss processes. This object later becomes a white dwarf (WD), and the former secondary companion, now the primary, presents in its photosphere vestiges of the mass accretion event. Jorissen & Boffin (1992) argue that, besides Roche lobe overflow, accretion from the strong wind of an AGB star can probably account for the observed heavy-element enrichment of the barium stars; they further assert that the efficiency of wind accretion does not depend on the stellar radius of the accreting star. Therefore, a key prediction of the wind mass transfer scenario is the existence of barium dwarf stars. A number of near–main-sequence F- and G-type stars has indeed been found to present excesses of the *s*-process elements with respect to Fe, less dramatic, however, on average, than found in the classical barium giant stars. Jorissen & Boffin (1992) review the subject of the barium dwarfs and list the limited number of known objects of this class. To this list were added the six barium dwarfs, plus four additional probable ones, uncovered by the large F star abundance survey of Edvardsson et al. (1993, hereafter E93), and also the eight barium dwarfs identified by North, Berthet, & Lanz (1994) among a sample of 20 F str λ4077 stars (F-type stars with strong Sr λ4077 lines).

In this Letter we report the discovery of the G5 V star HR 6094 as a new barium dwarf star. The observed overabundance of *s*-process elements in this object matches those found in the other barium dwarfs identified by E93. The quite distinctive characteristics of HR 6094 are its solar metallicity and the fact that it is the first near–zero-age barium dwarf star identified so far. This provides evidence that the barium phenomenon is not restricted to more or less old, metal-poor systems as proposed by most. Furthermore, the presence of a common proper motion WD companion 0.026 pc away suggests that the culprit of the *s*-process enhancement of a barium dwarf may have been directly spotted for the first time.

2. OBSERVATIONS AND ANALYSIS

The discovery of the abundance anomalies of HR 6094 results from an ongoing survey of the detailed abundance pattern of a selected sample of nearby solar-type stars. The Cassegrain echelle spectrograph of the CTIO 4 m telescope was used to obtain $R = 29{,}000$ resolution spectra, ranging from λ4500 (S/N > 300) to λ6450 (S/N > 1000), for 15 stars, including HR 6094, in 1994 March. In addition, the Hα region was observed at $R = 20{,}000$ and S/N = 200 with the coudé

[1] Observations collected at the Cerro Tololo Inter-American Observatory (CTIO), Chile, and at the Observatório do Pico dos Dias, operated by the CNPq/Laboratório Nacional de Astrofísica, Brazil.

TABLE 1
ELEMENT ABUNDANCES OF HR 6094

Element	Abundance	Species	Number of Lines
[Fe/H]........	+0.04	Fe I, Fe II	90, 9
[C/Fe]........	−0.18	C I	2
[Mg/Fe]......	−0.06	Mg I	2
[Si/Fe]........	−0.04	Si I	11
[Ca/Fe].......	0.03	Ca I	2
[Sc/Fe]	0.05	Sc II	5
[Ti/Fe]	0.07	Ti I, Ti II	15, 5
[V/Fe]........	0.12	V I	7
[Cr/Fe].......	−0.02	Cr I	12
[Mn/Fe]......	−0.07	Mn I	6
[Co/Fe].......	−0.03	Co I	9
[Ni/Fe].......	−0.06	Ni I	18
[Y/Fe]........	0.22	Y II	5
[Zr/Fe].......	0.24	Zr II	1
[Ba/Fe].......	0.37	Ba II	2
[Ce/Fe].......	0.06	Ce II	2
[Pr/Fe]	0.15	Pr II	1
[Nd/Fe]	0.27	Nd II	2
[Gd/Fe]	0.04	Gd II	1

spectrograph of the 1.60 m telescope of the CNPq/Observatório do Pico dos Dias, Brazil.

The equivalent widths of the HR 6094 spectra have been converted to elemental abundances using the MARCS models (Gustafsson et al. 1975), modified as described by E93, and a computer code kindly provided by M. Spite (Paris-Meudon Observatory). Solar *gf*-values were derived from equivalent widths measured off a spectrum of Ganymede, obtained with the same instrument, and a solar model atmosphere computed in the same code as that used for the star. The effective temperature ($T_{\rm eff}$) and surface gravity (log g) were derived from the excitation and ionization equilibria of a set of 90 Fe I and nine Fe II lines. The microturbulence velocity was obtained by forcing the abundances of the individual Fe I lines to be independent of line strength. Both the $T_{\rm eff}$ value derived by fitting theoretical profiles to the observed Hα profile (calculated as described by Praderie 1967) and the one inferred from the calibration of the $(B - V)$, $(b - y)$, and β color indices (Saxner & Hammarbäck 1985) show excellent agreement with the $T_{\rm eff}$ derived from the excitation and ionization equilibria.

A fully detailed description of the spectroscopic analysis and first abundance results for the above sample of nearby solar-type stars will be presented by Porto de Mello & da Silva (1997). Suffice it here to say that the analysis was rigorously differential, and that for the Sun, used as the standard star, we adopted $T_{\rm eff} = 5777$ K, log $g = 4.44$, and $\xi = 1.52$ km s^{-1}. The derived atmospheric parameters of HR 6094 are $T_{\rm eff} = 5859$ K, log $g = 4.63$, $\xi = 1.63$ km s^{-1} and [Fe/H] $= +0.04$ (as usual, $[A] = \log A – \log A_{\odot}$). The mean *internal* uncertainties in the atmospheric parameters are, respectively, for $T_{\rm eff}$, log g, [Fe/H], and ξ, 30 K, 0.12 dex, 0.06 dex, and 0.04 km s^{-1}. The mean uncertainty of the [element/Fe] ratios is 0.06 dex: this includes the uncertainties in the atmospheric parameters, in the solar *gf*-values, and in the observed equivalent widths. The abundances of the studied elements in HR 6094 are given in Table 1. For each of the heavy elements analyzed, with the exception of Ce, HR 6094 shows a marked overabundance with respect to Fe. According to Cameron (1982) and Raiteri, Gallino, & Busso (1992), the abundances of these elements are dominated by the *s*-process. However, the sole *r*-process–dominated element analyzed by us, Gd, is not overabundant, but we note that its abundance ratio is derived from one line only.

3. DISCUSSION

E93 discuss the six barium dwarf stars found in their survey along with four other candidates to such status. The [Ba/Fe] excesses of these objects range from +0.1 to +0.5: four of them present [Ba/Fe] $\geq +0.30$ and may be considered as reliably established barium dwarfs. It is interesting to note that these four stars have $-0.50 \leq$ [Fe/H] ≤ -0.10, similar to the metallicity of classical barium giants. For these stars, when data on Y, Zr, and Nd are available, they invariably confirm the overabundance of Ba. North et al. (1994) give Sr, Y, Zr, Ba, Ce, and Nd abundances of eight barium dwarfs, with $-0.9 <$ [Fe/H] < -0.3. For these objects, higher mean excesses of the *s*-process elements are found, ranging from +0.6 to +1.0, more similar to those of the classical barium giant stars.

The ages estimated by E93 for the four most likely barium dwarfs, from the $\delta c_1(\beta)$ index, range from 2 to 13 Gyr. Both E93 and North et al. (1994) suggest that the barium dwarfs are composed of more or less old, metal-poor, thick-disk stars. Jorissen & Boffin (1992) argue that barium stars are indeed formed almost exclusively in metal-poor systems, one possible reason being that metal-poor AGB stars become more easily enriched in *s*-process elements, another being the faster AGB winds encountered by higher metallicity AGB stars, which could lower the efficiency of wind accretion. North et al. (1994) and Smith et al. (1996), among other authors, present evidence that the neutron exposure is inversely correlated with metallicity, which might imply that the barium phenomenon occurs preferably in metal-poor systems. Such an inverse correlation is expected if the neutron source for the *s*-process is ^{13}C(α, n)^{16}O (Clayton 1988). As pointed by Luck & Bond (1991), the neutron exposure undergone by the accreted matter may be gauged qualitatively by the ratio of the mean abundance of the "heavy" *s*-process (*hs*) elements, Ba–Gd, to the "light" ones (*ls*), Sr, Y, and Zr. North et al. (1994) present a [*hs*/*ls*] abundance ratio versus [Fe/H] diagram for a sample of Ba giants and F str λ4077 stars (their Fig. 7), and a clear inverse correlation is seen. The abundance data of HR 6094 fit well in this scenario, and the moderate neutron exposure implied by its [*hs*/*ls*] $= -0.05$ ratio provides evidence that the neutron capture process does become less efficient at solar metallicity.

The abundance pattern of [*s*/Fe] of HR 6094 implies that it is a quite typical barium dwarf. A number of interesting peculiarities, however, distinguish HR 6094 from the other barium dwarfs so far identified. It is a near-ZAMS star, as revealed by its value of $T_{\rm eff} = 5859$ K plus the log $(L/L_{\odot}) =$ 0.14 luminosity and the resultant position in the theoretical H-R diagram of Charbonnel et al. (1993). The parallax error bar (van Altena, Lee, & Hoffleit 1995) puts its age as marginally consistent with age zero, a solar mass being inferred. Its high surface gravity, Ca II K core emission (Zarro & Rodgers 1983), and Li abundance log N(Li) = 1.99 (Soderblom et al. 1993) support strongly the notion that HR 6094 is a young star. In addition, Soderblom & Mayor (1993) establish HR 6094 as one of the probable members of the 0.3 Gyr old Ursa Major kinematical group, on the basis of space motions and chromospheric age indicators. Also, HR 6094 is a solar-metallicity star, being more metal-rich than any other barium dwarf identified to date.

Last, HR 6094 possesses, as a common proper motion companion, the DA white dwarf CD −38°10980, 5 mag fainter, located 5$'\!.$7 (∼0.026 pc) away. This degenerate companion is one of the nearest and brightest WDs and has been submitted to a number of detailed studies. The two stars clearly share the same motion in space (Wegner 1978), and their differential radial velocity (Koester 1987) is fully consistent with the expected gravitational redshift of CD −38°10980 (Holberg, Bruhweiler, & Andersen 1995). The present separation of HR 6094 and its companion is 5360 AU. Holberg et al. (1985, 1995) merge ultraviolet and optical data in a model atmosphere comprehensive study of its spectrum. Placing its atmospheric parameters into the evolutionary sequence of the 0.6 $M_\odot$ model of Koester & Schönberner (1986), a consistent age of about 30 Myr is derived. Considering 0.3 Gyr as the most probable age of the UMa group (Soderblom & Mayor 1993), a WD age of about 30 Myr would correspond to about 0.27 Gyr spent from the main sequence to the WD branch. Subtracting 10% of this time as having been spent on the giant branch, there is 0.24 Gyr left for the main-sequence lifetime, roughly corresponding to a 2.6 $M_\odot$ star. This mass value puts the former progenitor close to the mass interval where AGB stars are thought to contribute the most to the Galactic synthesis of the *s*-process elements (e.g., Busso et al. 1995).

Vedder et al. (1993) have observed CD −38°10980 with the *EUVE* satellite, detecting HR 6094, serendipitously, as an ultraviolet source. The WD is appreciably brighter than its solar-type companion in the UV, and therefore the presence of another, unresolved WD of about the same age next to HR 6094 can be ruled out. A much older, fainter and cooler WD could possibly pass undetected. However, Holberg et al. (1995) have observed HR 6094 with CORAVEL; their observations extended from 1985 to 1990 with a total time span of 1745 days. The radial velocity was found to be constant within 0.2 km s^{-1}, in the system of faint IAU standard stars, and no evidence of variability on timescales from days or weeks to 5 years was seen. In brief, the membership of the two stars in the young UMa group, the UV detection of CD −38°10980 as a much brighter source than HR 6094, and the reliably established constancy of the radial velocity of HR 6094 make the presence of another unseen WD next to HR 6094 highly unlikely.

Adding to the intriguing binary status of HR 6094 and its companion is the fact that E93 mention *another* UMa group member, the G0 V star HR 2047, as a possible barium dwarf candidate. The atmospheric parameters given by E93 for HR 2047 are $T_{\rm eff}$ = 5953 K, log g = 4.46 and [⟨Fe I, II⟩/H] = +0.01, and the heavy-element excesses are [Ba/Fe] = +0.25 and [Y/Fe] = +0.31. E93 compare the Ba overabundance of HR 2047 with the [Ba/Fe] versus age relation of the young stars of their sample and conclude that it could be considered as normal, in light of a possible inverse [Ba/Fe] correlation with age. If one examines the [Ba/Fe] versus [Fe/H] data of E93, however (their Fig. 15*k*), one sees that HR 6094 lies distinctly above the locus defined by the other [Fe/H] ∼ 0 stars, similar to the other barium dwarfs discussed by E93. Furthermore, the fact that E93 do not obtain ages for stars with less than about 1.6 Gyr means that one does not really know what is going on with [Ba/Fe] for stars with less than 0.5 Gyr. Porto de Mello & da Silva (1997) obtained [*s*/Fe] abundances for 14 nearby, field G-type stars: three of these (HR 3862, HR 4903, and HR 5011) are found to be young as revealed by chromospheric activity indicators and rotational velocity data. Their mean [*s*/Fe] value is −0.01 ± 0.09. The abundance pattern of HR 6094 shows it to be clearly overabundant in the heavy elements when compared to these young G-type stars. The case for HR 2047 is admittedly less strong owing to its lower [Ba/Fe], which might be compatible with the high-abundance tail of the [Ba/Fe] distribution of the normal stars. We have not found any recent Ba abundance determination for another UMa group member G-type star, but Oinas (1974) performed a model atmosphere analysis of ξ Boo (a "possible" UMa group member according to Soderblom & Mayor 1993), obtaining [Ba/⟨Fe I, Fe II⟩] = +0.2, based on one photographic plate, and therefore with very large uncertainty. In brief, existing data on the [Ba/Fe] abundance ratios of the UMa group stars argue in favor of the reality of the *s*-process enrichment of HR 6094, while HR 2047 might be regarded as a marginal case. Interestingly, the [*hs*/*ls*] ratio of the Ba and Y abundances of HR 2047 is −0.06, perfectly consistent with that of HR 6094.

HR 2047 is an astrometric binary with a 0.15 $M_\odot$ companion (Irwin, Yang, & Walker 1992) in a 14.2 yr orbit, probably an M dwarf, located 24 pc away from the HR 6094/WD pair (as given by the parallaxes of van Altena et al. 1995) and 16 pc away from the Sun. If HR 2047 is accepted as a marginal barium dwarf, the excess of heavy elements in its photosphere could be explained by an unseen WD companion. Radial velocity monitoring of HR 2047 by Irwin et al. (1992), with ∼20 m s^{-1} precision, suggests that the observed variation refers exclusively to the astrometric M dwarf companion; this has been confirmed over a 10 yr period by Walker et al. (1995). Any possible unseen WD companion would be very likely more massive than the M dwarf and influence the orbit revealingly. Also, this WD would probably be brighter than CD −38°10980 owing to the smaller distance and, if of similar age and luminosity, clearly visible in the UV wavelengths. In the second *EUVE* catalog (Bowyer et al. 1996), the detected HR 2047 flux in the 100 Å passband is comparable to that of ξ Boo A and B and π^1 UMa, UMa group members of comparable $T_{\rm eff}$, distance, and activity level. The flux level of these three stars is much lower than that measured for CD −38°10980 by these same authors. Dynamical and UV flux data, therefore, rule out the presence of a similar, unresolved WD close to HR 2047. Could these four stars, then, have formed a gravitationally bound system before CD −38°10980 underwent its major mass-loss episodes? Dynamical theory asserts that disruption of a bound system is more likely for multiple systems than for double ones. Perhaps HR 6094 and the WD formed an inner pair, around which HR 2047 and its M dwarf companion orbited at a greater distance.

In the classical barium star scenario, the TP-AGB star enriches its envelope with carbon, produced in the form of ^{12}C by shell He burning, in addition to the neutron capture elements. The classical barium giant stars do indeed show C overabundances, as do all the F str λ4077 stars discussed by North et al. (1994). A study of the C abundances of 105 disk F- and G-type stars was undertaken by Tomkin, Woolf, & Lambert (1995), and the metallicities of the studied stars span the −0.8 ≤ [Fe/H] ≤ +0.2 interval. One of the six E93 barium dwarfs is included in the Tomkin et al. (1995) sample, and this star does not present C enrichment. While we must await the analysis of the C abundances of the remaining five barium dwarfs of the E93 sample, the available data imply that the barium dwarfs found by the F star survey of E93 differ from the other *s*-process–rich stars by showing normal C abundances. Such a separation between the barium giant plus the F str λ4077 stars on one side, and the E93 barium dwarfs on another, was already suggested by the lower *s*-process overabundances of the latter group. HR 6094 does not conflict with this scenario, by being in fact C-*deficient* with [C/Fe] = −0.18 (Table 1). This deficiency might be attributable to the phenomenon known as "envelope burning" (Sackmann & Boothroyd 1992; Smith, Plez, & Lambert 1995; and references therein). This is the mechanism by which AGB stars reconvert themselves from C stars to O-rich S stars, by burning ^{12}C into ^{14}N, reducing the $^{12}C/^{13}C$ ratio and producing Li at the base of a"hot-bottomed" deep convective envelope when temperatures are in excess of about 20×10^6 K.

Concerning the hypothesis that the late AGB star, which contaminated the photospheres of HR 6094 and HR 2047 and now survives as the white dwarf CD −38°10980, underwent its major mass-loss episodes as a reconverted S star, whose envelope would be rich in the *s*-process elements and deficient in C, it is remarkable to note that *both* HR 2047 and HR 6094 are C-deficient by the same amount. Tomkin et al. (1995) obtained [C/Fe] = −0.22 for HR 2047. In the analysis of nearby G dwarfs of Porto de Mello & da Silva (1997), the three chromospherically active stars mentioned above have a mean [C/Fe] = −0.03 ± 0.08. Tomkin et al. (1995) find that for the [Fe/H] = 0 stars the mean value is [C/Fe] = 0. In fact, HR 2047 is the third most C-deficient star among their stars with −0.2 ≤ [Fe/H] ≤ +0.2. Such marked C deficiencies appear to be quite uncommon among solar-metallicity stars. According to Friel & Boesgaard (1990), the mean [C/Fe] value for the UMa group is −0.03: other young stellar groups such as the Pleiades and Hyades also present [C/Fe] ∼ 0. This strengthens our interpretation that CD −38°10980, HR 6094, and HR 2047 and its M dwarf companion formed a quadruple star system (probably hierarchical) up to ∼30 Myr ago, and the presently observed WD is the remnant of a late AGB star, which suffered envelope burning during its major mass-loss episode, reconverting itself from a C to an S star, thereby contaminating its dwarf companion stars with matter enriched in the *s*-process elements and depleted in C. This scenario is supported by the results of Smith et al. (1995), who found that the onset of envelope burning coincides with the very high luminosities prevailing at the end of the AGB stars' lives. These are terminated by a severe mass-loss episode, and thus most of the mass accreted by the companion would be *s*-process rich and C-deficient. We note, however, that the Li abundances of HR 2047 and HR 6094 are not distinctive among other UMa group probable members of similar T_{eff} (see Soderblom et al. 1993, their Fig. 3). It would be fruitful to perform a homogeneous detailed analysis of the abundances of Li, C, and the *s*-process elements for a sample of UMa group members: this could clarify decisively the status of the peculiar abundance patterns of HR 2047 and HR 6094 as well as possibly provide constraints to the theoretical models of AGB nucleosynthesis.

4. CONCLUSIONS

The young solar-type dwarf star HR 6094, member of the solar metallicity, 0.3 Gyr old, Ursa Major moving group, is found to be a barium dwarf, enriched in the *s*-process elements and deficient in C. HR 6094 is followed by the $V = 10$ common proper motion white dwarf CD −38°10980, 30 Myr old and located 5360 AU away, which is identified as the AGB star which enriched itself in the neutron capture elements and contaminated the photosphere of its companion via strong wind-driven mass loss. The existence of another *s*-process–enriched *and* C-deficient star in the UMa group, the G0 V dwarf HR 2047, forming an astrometric binary with, probably, an M dwarf, supports the notion that these four stars were gravitationally bound as a quadruple system in the past. This scenario is supported by the very similar [*s*/Fe] and [C/Fe] abundances of the two G dwarfs, and the age, T_{eff}, and luminosity of the WD. This finding implies that barium systems do not form exclusively in more or less old, metal-deficient systems. The observed C deficiency of the two barium dwarfs is ascribed to the onset of envelope burning in a hot-bottomed convective envelope in the late AGB star, which may have reconverted it from a C star to an S star, burning ^{12}C into ^{14}N and lowering the $^{12}C/^{13}C$ ratio. This may be the first direct detection of the degenerate component of a barium system.

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